SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 3

ERROR ANALYSIS TASK

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Course Code	ECA0908	**Course Name**	Digital Signal Processing

Assessment Tool Number	CO3-AT3
Assessment Tool Name	Error Analysis Task

Assigned Problem

Q20. A polyphase FIR filter gives shifted output samples because the coefficient indexing begins at phase 1 instead of phase 0. Debug the implementation. (10 Marks) (BL4) (WK4)

Solution – Error Analysis and Debugging

1. Background: Polyphase Decomposition

A polyphase FIR filter of length N, decimated/interpolated by factor M, splits the filter coefficients h[n] into M sub-filters called phases:

$$h\_k[n] = h[nM + k], \quad k = 0, 1, 2, \ldots, M - 1$$

Each phase k picks out every M-th coefficient starting at offset k. The output is reconstructed by combining the outputs of all M polyphase branches, each operating at the lower (decimated) rate. Phase 0 must contain h[0] — the tap that aligns with the current (undelayed) input sample.

2. Correct Phase Assignment (Index Origin = 0), Example M = 4

Phase k	Coefficients Picked
0	h[0], h[4], h[8], …
1	h[1], h[5], h[9], …
2	h[2], h[6], h[10], …
3	h[3], h[7], h[11], …

3. The Bug

The buggy implementation starts the phase index at 1 instead of 0:

```
for (k = 1; k <= M; k++) {         // WRONG: k starts at 1
   for (n = 0; n < N/M; n++) {
      h_poly[k][n] = h[n*M + k];  // shifts every phase by one tap
   }  }
```

Effect of the bug:

- Phase 1 (meant to hold h[0], h[4], h[8], …) now incorrectly holds h[1], h[5], h[9], …
- Phase 0, which should hold h[0], h[4], h[8], …, is never computed — array index 0 is left unused / uninitialized.
- Every branch effectively uses the next tap's coefficient set, and the branch that should carry h[0] (zero delay) is missing entirely.

Consequence: Since h[0] is never multiplied with the current (zero-delay) sample, the entire filter output is shifted by one sample period — a constant group delay error of 1 sample (or 1/M of a sample interval in the interpolator case). In a multirate system this also produces a periodic phase jump because the commutator switching order no longer matches the branch content.

4. Root Cause Analysis (Bloom Level 4 – Analyze)

Symptom	Cause
Output shifted by 1 sample	Loop/array indexing off-by-one (k = 1..M instead of 0..M−1)
Phase 0 branch empty / garbage	Index 0 of h_poly[] is never written to
Aliased/incorrect reconstruction in multirate use	Commutator expects branch order 0,1,…,M−1; mismatch corrupts synchronization

This is a classic off-by-one indexing error, typically arising when 1-based mathematical notation ("phases 1 to M") is translated directly into 0-based array code without adjustment.

5. Corrected Implementation

```
for (k = 0; k < M; k++) {            // FIX: k starts at 0
   for (n = 0; n < N/M; n++) {
      h_poly[k][n] = h[n*M + k];  // phase 0 correctly gets h[0], h[M], h[2M]…
   }  }
```

The commutator / output-combination loop must use the same 0-based ordering:

```
for (k = 0; k < M; k++) {
   y_poly[k] = 0;
   for (n = 0; n < N/M; n++)
      y_poly[k] += h_poly[k][n] * x_delay[k][n];  }
```

6. Verification

- Apply an impulse δ[n] to the corrected filter — the output should appear at n = 0, matching h[0], with no extraneous delay.
- Compare the group delay of the polyphase output with the direct-form FIR: both should show delay = $(N - 1)/2$ samples, not $(N - 1)/2 + 1$.
- Confirm that the phase-0 branch coefficient array now equals {h[0], h[M], h[2M], …} exactly.

7. Justification with DSP Concepts

Polyphase decomposition relies on the strict correspondence between phase index k and the modulo-M position of each coefficient in the original filter, $h_k[n] = h[nM + k]$. Any offset in k breaks this correspondence and is mathematically equivalent to convolving the input with a version of h[n] delayed by one sample, i.e. $h[n - 1]$. By the time-shifting property of the Z-transform, $H(z) \rightarrow z^{-1}H(z)$, which explains the uniformly shifted output. Re-indexing k from 0 to $M - 1$ restores the correct correspondence between commutator branch and filter tap, eliminating the spurious delay while preserving the frequency response and overall filter length N.

8. Summary

The 1-sample output shift was caused by an off-by-one error in phase indexing (loop starting at $k = 1$ instead of $k = 0$), which excluded h[0] from any branch and misaligned every subsequent coefficient-to-branch mapping. The fix is to index phases from 0 to $M - 1$, ensuring phase 0 always contains h[0], thereby restoring correct time alignment and group delay in the polyphase FIR structure.